Student Answer Sheet

Subject: Artificial Intelligence

Definition:

Artificial Intelligence (AI) is the field of computer science dedicated to creating systems that can perform tasks requiring human intelligence. It includes processes like learning, reasoning, problem-solving, and self-correction. AI systems can perceive their environment and take actions to maximise their chance of achieving goals, enabling machines to handle tasks such as visual perception, speech recognition, decision-making, and natural language understanding.

Body:

AI encompasses several subfields. Machine Learning allows systems to learn from data without explicit programming. Deep Learning uses multi-layered neural networks to recognise complex patterns in images, speech, and text. Natural Language Processing (NLP) enables machines to understand and generate human language, powering chatbots and translation tools. Computer Vision interprets visual data for applications like face recognition and autonomous vehicles. Core techniques include supervised learning (labelled data), unsupervised learning (unlabelled data), and reinforcement learning (reward-based). AI is transforming healthcare (diagnostics), finance (fraud detection), transportation (self-driving cars), and education (personalised learning).

Conclusion:

AI is a transformative technology reshaping industries globally. Its benefits include automation, improved efficiency, and enhanced problem-solving capabilities. However, it also presents challenges: ethical concerns around privacy, algorithmic bias, and job displacement require careful consideration. Responsible AI development must ensure transparency, fairness, and accountability. Future research in explainable AI, artificial general intelligence, and quantum AI will determine the long-term impact of this technology on humanity.